Here's how it works: Inside the computer, the memory stores two complex numbers, α_u and α_d, subject to the usual normalization rule,

$$\alpha_u^* \alpha_u + \alpha_d^* \alpha_d = 1.$$

At the beginning of the game, the α coefficients are initialized at some value. The computer then solves the Schrödinger equation to update the α's exactly as if they were the components of the spin's state-vector.

The computer also stores the classical three-dimensional orientation of the apparatus in the form of two angles or a unit vector. The keyboard allows you to set these angles and change them at will. One more element is stored in the memory, namely the value (either +1 or −1) representing the number in the window of the apparatus. The computer screen shows the apparatus. As the experimenter, you get to choose how your apparatus will be oriented. There is also a measure button M that activates the apparatus.

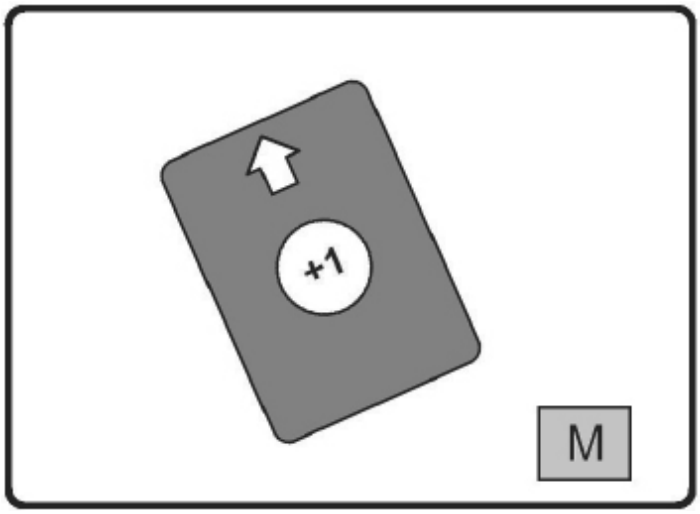


Figure 7.1: Quantum Sim. The computer screen displays the user-controlled orientation of the apparatus. For simplicity, only the two-dimensional orientation is shown here. The user can press the M button whenever she or he wants to measure the spin (not shown). Between measurements, the spin state evolves according to the Schrödinger equation.

The final element of the program is a random number generator that produces the measurement results +1 or −1 with probabilities $\alpha^*_u \alpha_u$ and $\alpha^*_d \alpha_d$, respectively. Keep in mind that random number generators are not really